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Project Overview

- Goal: predict whether a room is occupied using environmental sensor data.
- Dataset: UCI Occupancy Detection dataset.
- Task type: supervised binary classification.
- Models compared:
  - Logistic Regression
  - Random Forest
- Main result: scaled Logistic Regression performed best overall.

Why Occupancy Detection Matters

- Smart buildings need reliable occupancy estimates.
- Occupancy information can improve:
  - lighting control
  - heating and cooling
  - ventilation
  - energy efficiency
- Environmental sensors are less invasive than cameras.
- This makes the problem useful for offices, schools, and homes.

Dataset

- Source: UCI Machine Learning Repository.
- Total observations: 20,560.
- Target variable: `Occupancy`.
- Sensor features:
  - Temperature
  - Humidity
  - Light
  - CO_2
  - HumidityRatio
- The timestamp column was parsed but not used as a model feature.

Dataset Splits

Table 1: Predefined Dataset Splits

Split	Rows
Training	8,143
Validation	2,665
Final holdout test	9,752

The validation split was used for model comparison. The final holdout test split was used only for final evaluation.

Class Balance

Class Balance Counts

Table 2: Occupancy Counts by Split

Split	Not Occupied	Occupied
Training	6,414	1,729
Validation	1,693	972
Test	7,703	2,049

Because the dataset is imbalanced, F1 score, precision, and recall are more informative than accuracy alone.

Preprocessing

- The first column in each data file was treated as a row index.
- No missing values were found.
- The same five sensor features were used for both algorithms.
- Logistic Regression was tested with and without feature scaling.
- Scaling was done inside a scikit-learn pipeline to avoid data leakage.

Modeling Approach

Logistic Regression

- Simple baseline model.
- Fast to train.
- Easy to interpret.
- Works well when classes are close to linearly separable.

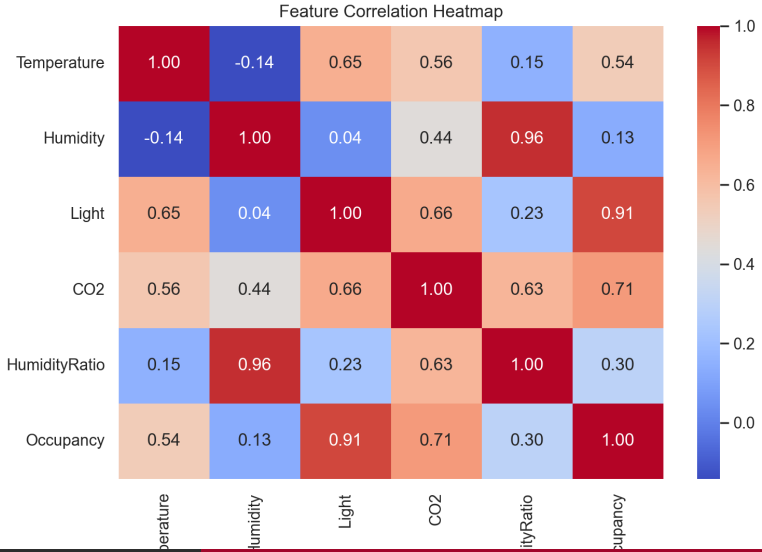
Random Forest

- Ensemble of decision trees.
- Can model nonlinear patterns.
- Provides feature importance.
- More complex than Logistic Regression.

Evaluation Metrics

- Accuracy: overall proportion of correct predictions.
- Precision: how many predicted occupied cases were actually occupied.
- Recall: how many actual occupied cases were correctly detected.
- F1 score: balance between precision and recall.
- Confusion matrix: shows false positives and false negatives directly.

Exploratory Data Analysis



EDA Takeaways

- Light has the strongest relationship with occupancy.
- CO_2 also has a meaningful relationship with occupancy.
- Feature scales differ significantly.
- Example: Light ranges from 0 to 1546.333, while HumidityRatio stays near 0.004.
- This supports testing feature scaling for Logistic Regression.

Validation Results

Table 3: Validation Metrics

Model	Accuracy	Precision	Recall	F1
Logistic Regression scaled	0.9771	0.9470	0.9928	0.9694
Logistic Regression unscaled	0.9764	0.9469	0.9907	0.9683
Random Forest	0.9501	0.9458	0.9156	0.9305

Scaled Logistic Regression had the best validation F1 score.

Validation Confusion Matrices

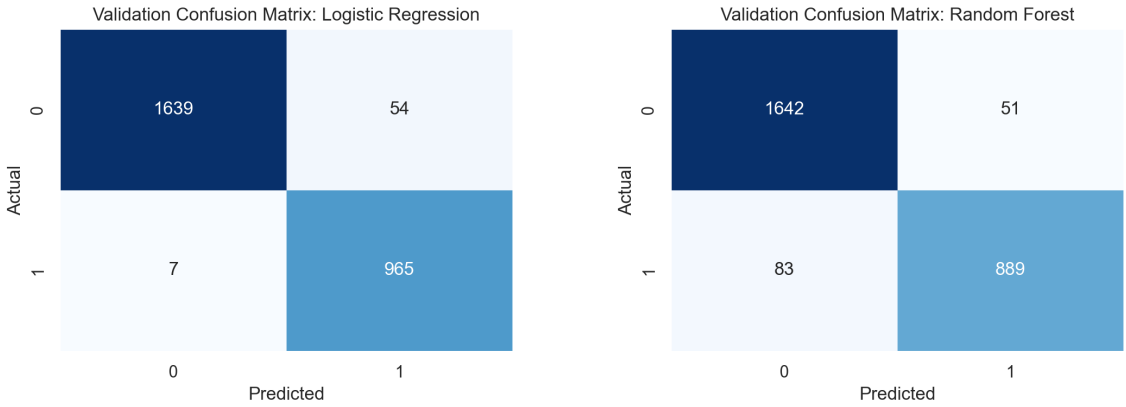


Figure 3: Validation confusion matrices: Logistic Regression and Random Forest.

Validation Interpretation

- Logistic Regression had much higher recall than Random Forest.
- Logistic Regression validation confusion matrix:
  - 54 false positives
  - 7 false negatives
- Random Forest validation confusion matrix:
  - 51 false positives
  - 82 false negatives
- Random Forest missed more occupied cases on the validation split.

Final Holdout Test Results

Table 4: Final Holdout Metrics

Model	Accuracy	Precision	Recall	F1
Logistic Regression scaled	0.9925	0.9718	0.9932	0.9824
Random Forest	0.9867	0.9436	0.9961	0.9691

The final test results confirmed that scaled Logistic Regression was the strongest overall model.

Holdout Confusion Matrices

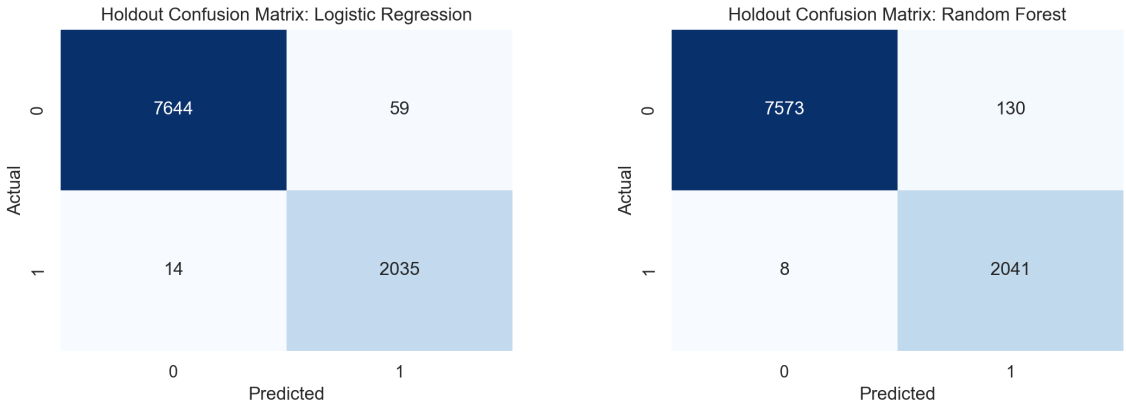
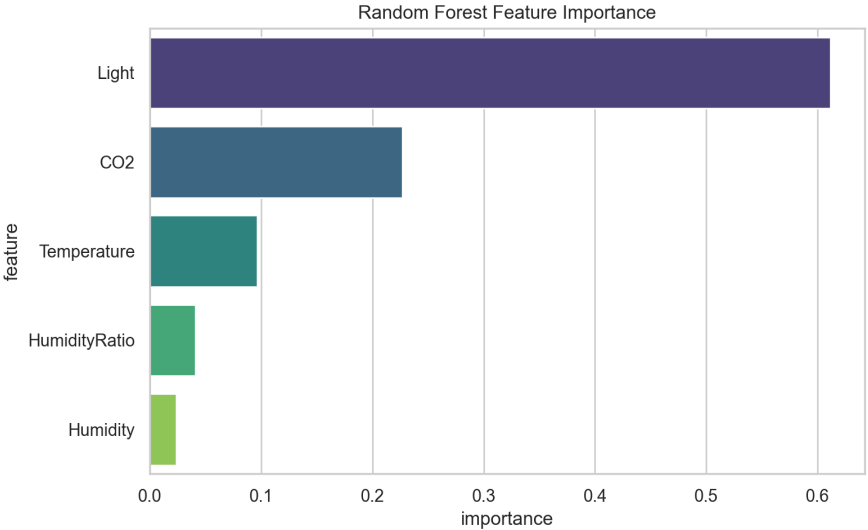


Figure 4: Holdout confusion matrices: Logistic Regression and Random Forest.

Holdout Interpretation

- Logistic Regression made fewer total errors on the holdout test set.
- Logistic Regression:
  - 59 false positives
  - 14 false negatives
- Random Forest:
  - 122 false positives
  - 8 false negatives
- Random Forest had slightly higher recall, but lower precision and lower F1.

Random Forest Feature Importance



Most Important Features

Table 5: Random Forest Feature Importance

Feature	Importance
Light	0.6127
CO_2	0.2272
Temperature	0.0962
HumidityRatio	0.0404
Humidity	0.0235

Light and CO_2 were the strongest predictors of occupancy.

Main Findings

- Scaled Logistic Regression was the best overall model.
- The final holdout F1 score was 0.9824.
- Random Forest performed well, but did not outperform Logistic Regression.
- Light and CO_2 were the most important sensor signals.
- A simple model was enough for this dataset.

Limitations

- Only two algorithms were compared.
- The timestamp was not engineered into time-based features.
- The results come from one dataset and one room environment.
- More testing would be needed before deploying the model in other buildings.

Future Work

- Add time-based features such as hour of day.
- Compare more algorithms, such as SVM or Gradient Boosting.
- Tune Random Forest hyperparameters more deeply.
- Test the model on other rooms or buildings.
- Explore real-time IoT deployment.

Conclusion

- Room occupancy can be predicted accurately using environmental sensor data.
- The best model was scaled Logistic Regression.
- Final holdout F1 score: **0.9824**.
- Light and CO_2 were the strongest indicators of occupancy.
- This supports non-invasive occupancy detection for smart-building applications.

References

1. Candanedo, L. M., & Feldheim, V. (2016). Accurate occupancy detection of an office room from light, temperature, humidity and CO_2 measurements using statistical learning models. *Energy and Buildings, 112*, 28–39.
2. UCI Machine Learning Repository. Occupancy Detection Data Set.
3. Zhang, W., Han, K., Costa, G., & Li, J. (2022). A review on occupancy prediction through machine learning for enhancing energy efficiency, air quality and thermal comfort in the built environment. *Renewable and Sustainable Energy Reviews*.

Questions?